

HARSHIT PAL

Full-Stack Developer & AI / IoT Systems Engineer

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HONORS, CERTIFICATIONS & NATIONAL QUALIFICATIONS

- **2x NDA Qualified & 3x SSB Conference Out:** Cleared National Defence Academy entrance exam twice and attended 3 Services Selection Boards (demonstrated national-level leadership, psychological resilience, and crisis decision-making).
- **IOQM 2022 Qualified:** Merit Certificate Holder in Indian Olympiad Qualifier in Mathematics (awarded for top national mathematical problem-solving talent).
- **AWS Certified AI Practitioner:** Industry-recognized certification for generative AI architectures, foundational models, cloud compliance, and ML pipeline development.
- **Innotech 2025 Hackathon Winner:** Awarded First-Year Innovator at KIET Ghaziabad for engineering functional IoT hardware mesh systems.
- **JEE Main 2025:** 89.12 Percentile in national level engineering entrance examination.

EDUCATION

KIET Group of Institutions — *B.Tech in CSE (Artificial Intelligence & Machine Learning)*

2025 – 2029 | SGPA: **7.83**

SD Public School — *Senior Secondary (Class XII: 81.2%) & High School (Class X: 94.6%)*

Muzaffarnagar, UP

TECHNICAL PROJECTS

BIS Standards Recommendation Engine | AI / Hybrid RAG Retrieval Engine

[Live Demo](#) | [GitHub](#)

- **Sub-0.01s Hybrid Search:** Engineered deterministic RAG system parsing Bureau of Indian Standards (BIS SP 21) cement/steel specifications for Indian MSEs using dense FAISS vector embeddings and sparse BM25 retrieval.
- **Benchmark Precision:** Achieved **100% Hit Rate @3**, **MRR @5 of 1.000**, and automated validation passing **59/59 robustness tests** with FastAPI and domain term expansions.

StudyShare Suite & Ingestion Automation | Next.js, Flutter, NestJS, Neon, Tesseract, n8n

studysshare.in | [GitHub](#)

- **Campus ERP Ecosystem:** Built live campus portal and mobile app deployed on Railway/Neon PostgreSQL serving **500+ active users** for leave requests, fee verification, and complaint dispatch.
- **Automated OCR Ingestion:** Shipped an n8n webhook pipeline integrating Tesseract OCR for automated fee receipt parsing (98.4% accuracy, 1.2s latency) and dynamic encrypted QR student gate pass verification.

Memori Labs | Agent-Native Memory Infrastructure (SQL + LLM State Layer)

memorilabs.ai | [GitHub](#)

- **State Management for Agents:** Architected an LLM-agnostic, SQL-native memory layer capturing agent execution graphs, tool outputs, and user sessions into queryable structured persistent state.

ReturnShield AI (Innogeeks Hackathon) | Python, scikit-learn, React 19, Vite, Edge

returnsheild-ai.vercel.app | [GitHub](#)

- **ML Return Risk Classifier:** Built an e-commerce fraud and return prevention system at Innogeeks Hackathon using trained scikit-learn classifiers with real-time order risk scoring (95.0% accuracy, 45ms inference) and revenue loss exposure estimation.
- **Interactive Merchant Intelligence:** Implemented a high-performance React 19 + Geist design dashboard with automated action stacks, 12-week risk trend visualization, and Vercel Edge Copilot.

ESP32 Smart Infrastructure & IoT Mesh Networks | C++, ESP-NOW, MQTT, FreeRTOS

[Wokwi Sim](#) | [Demo](#)

- **Zero-GSM Urban Mesh:** Developed low-power municipal manhole monitoring hardware utilizing ESP32, HX711 load cells, and ultrasonic sensors communicating over self-healing ESP-NOW peer-to-peer mesh relays.

TECHNICAL SKILLS & CORE COMPETENCIES

Languages:	TypeScript, JavaScript, Python, Dart, C++, C, SQL, HTML5/CSS3
Frontend & Mobile:	React, Next.js, Flutter, Tailwind CSS, shadcn/ui, Vite, Framer Motion
Backend & APIs:	Node.js, Express, NestJS, FastAPI (Python), Prisma ORM, RESTful APIs, WebSockets
AI, Cloud & Data:	FAISS Vector DB, Hybrid RAG, BM25, PostgreSQL, Neon Serverless, Redis, Docker, Vercel, Railway, Cloudflare, Linux, n8n
Hardware & IoT:	ESP32, ESP-NOW Mesh Protocol, Arduino, MQTT, FreeRTOS, Sensor Interfacing (PIR, Ultrasonic, Load Cells)